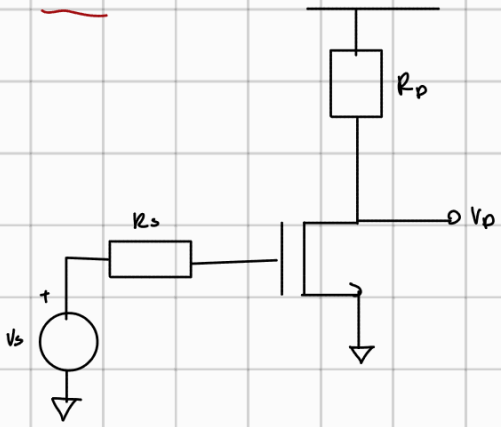
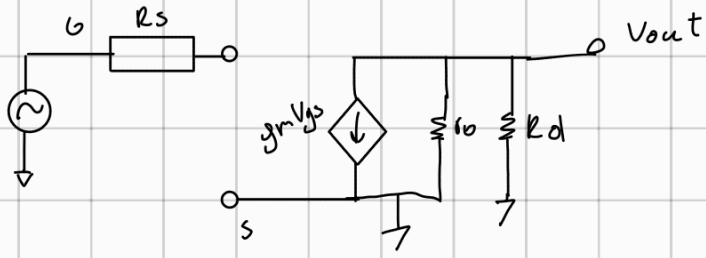
Ders6

Ex:



$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \qquad \frac{1}{4kTR_s} = ?$$

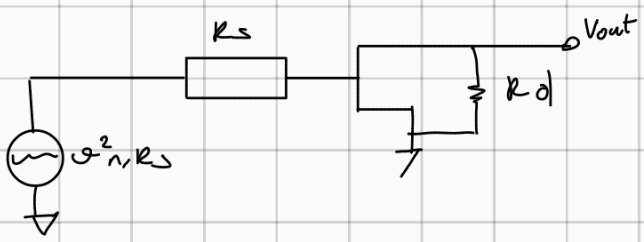
$$A_{vt} = \left(\frac{v_{out}}{v_{in}}\right) \cdot \frac{v_{in}}{v_s}$$



$$\approx -g_m \,(r_o \parallel R_d)$$

$$\frac{v_o}{v_i} = -g_m (r_o \parallel R_d) \approx -g_m R_d$$

$\overline{v_{n,out}^2}$ (total) → ① Rs ($v_s = 0$) →



$$\overline{v_{n,out_1}^2} = \overline{v_{n,Rs}^2} \underbrace{(-g_m r_o \parallel R_d)^2}_{\text{Transistor gain}} = 4kTR_s\,(g_m\,(r_o \parallel R_d))$$

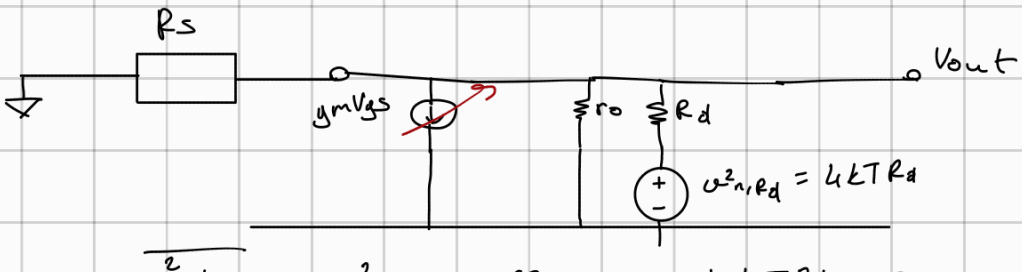
② Noise from mosfet



$$\overline{i_n^2} = 4kT\gamma g_m$$

$$\overline{v_{out,2}^2} = \overline{i_n^2} \cdot (r_o \parallel R_d)^2$$

③ Noise from Rd



$$\overline{v_{out,3}^2} = \overline{v_{n,Rd}^2} \cdot \frac{r_o}{r_o + R_d} = 4kTR_d \cdot \frac{r_o}{r_o + R_d}$$

$$v_{n,out}^2 = 4kTR_S \left(g_m (r_o \parallel R_d) \right) + \overline{i_n^2} (r_o \parallel R_\phi)^2 + 4kTR_d \cdot \frac{r_o}{r_o + R_d}$$

$$\frac{\frac{v_{n,out}^2}{A_{vt}^2}}{\frac{1}{4kTR_S}} = \frac{4kTR_S \left(g_m (r_o \parallel R_d) \right) + \overline{i_n^2} (r_o \parallel R_\phi)^2 + 4kTR_d \cdot \frac{r_o}{r_o + R_d}}{\left(g_m (r_o \parallel R_d) \right)^2 \cdot 4kTR_S}$$